

Two-Tier Skill Routing Architecture

User Query: "Add hover animations to button component"

Fast Path (Tag + Keyword)

Tokenize query -> Match tags/summary -> Score
Threshold: score $\geq$ 2 -> Return (<1ms)

score $\geq$ 2?
No Yes

TF-IDF Fallback

Cosine similarity over TF-IDF vectors
Return top-K results (~3ms)

Ranked Skill Results

1. animate (0.92) 2. frontend-design (0.87) 3. design-taste-frontend (0.81)

Load full SKILL.md on demand □ 97.2% token savings

Performance: 92% fast path, 8% TF-IDF fallback, 0.57ms avg latency

7.6K index tokens vs 273K full-body tokens (97.2% savings)